CURRICULUM VITAE

NAME: Luz América Chi Uluac (L. América Chi)	**ORCID ID:** 0000-0002-2606-0357
POSITION TITLE: Senior Postdoctoral Researcher	**EMAIL:** achi@uidaho.edu
PRIMARY ORGANIZATION & LOCATION: Department of Physics, University of Idaho, Moscow, Idaho, USA	**GitHub:** AmericaChi **Google Scholar:** L. América Chi

PERSONAL STATEMENT

Physical engineer with a **PhD in Physical Chemistry** and a computational scientist who designs molecular-level models to tackle multidisciplinary challenges across biology, physics, chemistry, and materials science. I have developed **neural-network** models that substantially **improve binding free-energy predictions** for single and multiple protein mutations; built **AI-based structural models** that identify mutations driving host restriction or viral escape, **confirmed in viral mini-replicon assays**; identified candidate α5β1-integrin ligands that supported a **provisional patent** filing; and assessed an MD + in-silico mutagenesis pipeline for predicting antibody-escape mutations with **strong agreement to experimental data**. I currently contribute to a **U.S. Department of Defense** project on the computational design of antimalarial compounds and in the structural arm of the dual-framework to identify Ubiquitin Proteasome System proteins in *P. falciparum*. I have been a member of the drug discovery group at the University of Idaho since 2023, working on antifungal candidates against potato pathogens. I've been a regular **presenter** at the **Biophysical Society Annual Meeting** since 2024. I also **co-teached Biophysics** to **undergraduate and graduate** students in the spring of 2026. My experience spans **five years of postdoctoral** research across three **two countries** and **several institutes and departments**, **nine publications**, **four preprints** in different stages of peer-review and **several drafts in progress**, and a track record of **successful** competitive **CPU/GPU computing-time proposals** on HPC systems in **Mexico, Germany, and the USA.**

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	COMPLETION DATE	FIELD OF STUDY
Autonomous University of Yucatán (UADY), Mexico	BS	01/2013	Physical Engineering
Center for Scientific Research of Yucatán (CICY), Mexico	MSc (Summa Cum Laude)	03/2015	Polymeric Materials
CINVESTAV-IPN, Mexico	PhD	11/2020	Physical Chemistry
Superior School of Medicine (ESM), IPN, Mexico	Postdoctoral	02/2023	Biomolecular Simulations
IMCI, University of Idaho, USA	Postdoctoral	06/2024	Molecular Modeling
Dept. of Chemical & Biological Engineering, University of Idaho, USA	Postdoctoral	06/2025	Membrane Proteins

Department of Physics, University of Idaho, USA	Postdoctoral	Present	AI and Drug Design

- **PhD dissertation:** Biomolecular simulations, from peptide design to ion channels. Supervisor: Dr. María Cristina Vargas González & Professor Paolo Carloni.
- **MSc dissertation:** Bilayer scaffold via electrospinning for potential use in skin tissue engineering. Supervisor: Dr. Fernando Hernández Sánchez.
- **BS dissertation:** Nanomaterials for wastewater treatment. Supervisor: Dr. Roger Amílcar González Herrera.

PROFESSIONAL EXPERIENCE

Postdoctoral Fellow | Department of Physics, University of Idaho, USA *Development of antimalarial PROTACs. Supervisor: F. Marty Ytreberg.*	07/2025 – Present
Postdoctoral Fellow | Dept. of Chemical and Biological Engineering, University of Idaho, USA *Molecular mechanisms underlying the spectral shift in zebrafish cone opsins. Supervisor: Jagdish Patel.*	05/2024 – 06/2025
Postdoctoral Fellow | Institute for Modeling Collaboration and Innovation (IMCI), University of Idaho, USA *Prediction of viral antibody escape mutations to SARS-CoV-2. Supervisors: Craig Miller, F. Marty Ytreberg, and Jagdish Patel.*	02/2023 – 05/2024
Postdoctoral Fellow | Laboratory for the Design and Development of New Drugs and Biotechnological Innovation, ESM, IPN, CDMX, Mexico *Ang-(1-7)/PAMAM-dendrimer complexation and Ang-(1-7) oligomerization. Supervisor: Marlet Martínez Archundia.*	10/2021 – 02/2023
Online Course Developer | CINVESTAV–UNICEF, Mexico *High-school physics course supporting student learning during the COVID-19 pandemic.*	07/2021 – 12/2021

RESEARCH SUPPORT

U.S. Department of Defense (DOD) | Project 1I80VP000634-01 PI: F. Marty Ytreberg. Role: Postdoctoral Researcher. *Computational modeling and design of antimalarial compounds.*	06/2025 – Present

WORKING GROUPS

NNOVAH – Drug Discovery Group | University of Idaho, USA *Member. Computational discovery of antifungal compounds against potato pathogens.*	02/2023 – Present

HPC COMPUTING-TIME AWARDS

Competitive CPU/GPU computing-time proposals awarded to me as applicant over the years:

SYSTEM	INSTITUTION / LOCATION
Xiuhcoatl	CINVESTAV, Mexico
ABACUS	ABACUS Laboratory, CINVESTAV, Mexico
Kukulkán	CINVESTAV, Mexico
JURECA	Forschungszentrum Jülich (FZJ), Germany
Lemhi	USA
RCDS Core	Research Computing & Data Services, University of Idaho, USA

TECHNICAL SKILLS

Modeling	Molecular dynamics, molecular docking and virtual screening, **replica exchange,** Monte Carlo, **binding free-energy estimation**, **metadynamics and enhanced sampling**, computational mutational scanning, machine learning, bioinformatics, structural modeling.
Research lines	Peptide encapsulation on dendrimers, peptide oligomerization, protein-peptide binding, de novo peptide design, antibody escape prediction, binding free-energy prediction, GPCR mechanisms, Cl^- antiporter selectivity, protein-DNA interactions, targeted protein degradation, viral-antiviral interactions.
Software	GROMACS, VMD, PyMOL, MODELLER, Discovery Studio, LigPlot, AutoDock Vina, PLUMED, CHARMM-GUI, FoldX, Schrödinger Maestro, ChimeraX, pyDockDNA, AlphaFold, SWISS-MODEL, Jupyter, Inkscape, DALI.
Programming / HPC	Python, Bash scripting; HPC and Linux/Unix environments.
Languages	Spanish, English, French.

PUBLICATIONS

1. Barnes, J. E., **Chi, L. A**. (co-first author), Ytreberg, F. M., & Patel, J. S. (2026). Leveraging neural networks to correct FoldX free energy estimates. ACS Omega. https://doi.org/10.1021/acsomega.6c01409
2. Han, Z., Alkhadrawi, A., Tofani, S., Zhang, W., **Chi, L. A**., Maruwan, J., ... & Su, W. W. (2026). A constrained degron tag engineered to enhance auxin-inducible protein degradation. ACS Synthetic Biology. https://doi.org/10.1021/acssynbio.5c00970
3. Blanco-Rodriguez, R., **Chi, L. A**., & Hernandez-Vargas, E. (2026). Multicompartment modeling of influenza A replication along the murine respiratory tract. Frontiers in Virology. https://doi.org/10.3389/fviro.2026.1772993
4. **Chi, L. A**., Barnes, J. E., Patel, J. S., & Ytreberg, F. M. (2024). Exploring the ability of the MD+FoldX method to predict SARS-CoV-2 antibody escape mutations using large-scale data. Scientific Reports, 14, 23122. https://doi.org/10.1038/s41598-024-72491-z
5. **Chi-Uluac, L. A**., Asgharpour, S., Blanco-Rodríguez, R. G., & Martínez-Archundia, M. (2023). Atomistic molecular insights into angiotensin-(1-7) interpeptide interactions. Journal of Chemical Information and Modeling, 63(16), 5331–5340. https://doi.org/10.1021/acs.jcim.3c00478
6. **Chi, L. A**., Asgharpour, S., Correa-Basurto, J., Bandala, C. R., & Martínez-Archundia, M. (2022). Unveiling the G4-PAMAM capacity to bind and protect Ang-(1-7) bioactive peptide by molecular dynamics simulations. Journal of Computer-Aided Molecular Design, 36(9), 653–675. https://doi.org/10.1007/s10822-022-00470-5
7. Asgharpour, S., **Chi, L. A**., Spehr, M., Carloni, P., & Alfonso-Prieto, M. (2022). Fluoride transport and inhibition across CLC transporters. In Anion Channels and Transporters (pp. 81–100). Springer. https://doi.org/10.1007/164_2022_593
8. **Chi, L. A.**, & Vargas, M. C. (2020). In silico design of peptides as potential ligands to resistin. Journal of Molecular Modeling, 26, 1–14. https://doi.org/10.1007/s00894-020-4338-3
9. Cob-Calan, N. N., **Chi-Uluac**, L. A., Ortiz-Chi, F., Cerqueda-García, D., Navarrete-Vázquez, G., Ruiz-Sánchez, E., & Hernández-Núñez, E. (2019). Molecular docking and dynamics simulation of protein β-tubulin and antifungal cyclic lipopeptides. Molecules, 24(18), 3387. https://doi.org/10.3390/molecules24183387

PREPRINTS

1. **Chi, L. A.**, Levy, M., Geiger, R., Malik, H., & Patel, J. S. (2026). Structural modeling and experimental validation define the MxA–Thogotovirus nucleoprotein interface that drives restriction and escape. **bioRxiv**. https://doi.org/10.64898/2026.08.07.743570
2. **Chi, L. A.**, & Ytreberg, F. M. (2026). PepXPro: a framework for curating, generating, and optimizing structure-affinity protein-peptide datasets. **bioRxiv.** https://doi.org/10.64898/2026.08.09.743757
3. **Chi, L. A**., Pandey, S. K., Kolodziejczyk, W., Lund-Andersen, P., Barnes, J. E., Kapusta, K., & Patel, J. S. (2024). Molecular mechanisms underlying the spectral shift in zebrafish cone opsins. **bioRxiv**. https://doi.org/10.1101/2024.09.24.614827
4. **Uluac, L. A. C**., & González, M. C. V. (2021). Molecular dynamics simulations of resistin and RELMβ proteins: insight into structural dynamics. **Authorea Preprints**. https://doi.org/10.22541/au.163251023.36890937/v1

DRAFTS IN PROGRESS

1. Megan E. Levy, Rechel A. Geiger, L. America Chi, Peter Dietzen, Valentina Wagner, Michael Emerman, Jagdish Patel, Laura Graf, Harmit S. Malik (2026). Targeted Engineering of Human MxA Yields a Broad-Spectrum Super-Restrictor of Orthomyxoviruses, Including Influenza Viruses. Submitted to bioRxiv.
2. Rosales-Hernández, M. C., Guerrero González, I., **Chi Uluac, A**., Olvera Valdez, M., Ramírez-Salinas, G. L., & Martínez-Archundia, M. Revealing the binding mechanisms of loperamide and glipizide on tetraspanin CD151 and investigation of their antiproliferative activity on cancer cell lines. (Draft in progress)
3. Fortunato, E. **A., Chi, L. A**., & Patel, J. Modeling pp71–DNA interactions. (Draft in progress)
4. **Chi, L. A.**, Hendricks, S., & Ytreberg, F. M. A structural-bioinformatic framework for mapping the UPS of Plasmodium falciparum. (Draft in progress)

PATENTS | AWARDS | FELLOWSHIPS

Provisional Patent (WSU) | Chi, L. A., Patel, J., Myura, T., Bose, S., et al. Virtual screening strategy for α5β1-integrin ligands.	2025
University of Idaho Postdoctoral Association (IPA) Travel Award – $450 USD 70th Biophysical Society Annual Meeting, San Diego, CA.	2026
Biophysical Society Annual Meeting Travel Award – $350 USD 69th Biophysical Society Annual Meeting, Los Angeles, CA.	2025
DAAD Research Grant – Short-Term Grant (57440917) | €1,200/month + €1,800 travel allowance Structural basis of the (S to P) mutation in ClC-ec1 transporter by classical MD simulations. PI: Paolo Carloni. Forschungszentrum Jülich, Germany.	05–09/2019
MEXFITEC Scholarship | Bilateral International Mobility Program Mexico–France for Engineers	2009–2010

INVITED AND CONTRIBUTED TALKS

Platform talk – 70th Biophysical Society Annual Meeting, San Francisco, CA, USA *Leveraging neural networks to correct FoldX free energy estimates.*	Feb 2026
Guest lecture – Foundations of Machine Learning (Math 4830), University of Idaho *ML-driven molecular modeling.*	Dec 2025
UI External Advisory Committee visit, University of Idaho *Molecular mechanisms underlying the spectral shift in zebrafish cone opsins.*	Oct 2024
SIAM Conference, Spokane, WA, USA *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Jul 2024

78th Northwest Regional Meeting (NORM), Washington State University, Pullman, WA, USA *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Jun 2024
Shot of Science, UI Postdoc Association, University of Idaho *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Oct 2023
Anticipating the Evolution and Dynamics of Emerging Pathogen Threats Workshop, University of Texas at Austin, USA *Prediction of antibody escape mutations to SARS-CoV-2.*	Jun 2023
Eleventh Molecular Dynamics Workshop, Institute of Physical Sciences, UNAM, Mexico *Unveiling the G4-PAMAM capacity to bind and protect Ang-(1-7) bioactive peptide by MD simulations.*	Jul 2022
3rd DynIon Meeting, Max Planck Institute for Biophysical Chemistry, Göttingen, Germany *Biomolecular simulations on CLC ion channels.*	Jun 2019

POSTERS

Royal Society of Chemistry Poster Conference, online (LinkedIn) *Exploring the ability of the MD+FoldX method to predict SARS-CoV-2 antibody escape mutations using large-scale data.*	Mar 2025
69th Biophysical Society Annual Meeting, Los Angeles, CA, USA *Molecular mechanisms underlying the spectral shift in zebrafish cone opsins.*	Feb 2025
LatinXChem 2024, online poster conference *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Oct 2024
IHHE Annual Research Symposium, University of Idaho *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Apr 2024
68th Biophysical Society Annual Meeting, Philadelphia, PA, USA *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Feb 2024
LatinXChem 2023, online poster conference *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Oct 2023
UI External Advisory Committee poster session, University of Idaho *Exploring the accuracy of SARS-CoV-2 antibody escape predictions by using extensive large-scale data.*	Sep 2023

Fourth Nanotechnology Symposium, Universidad Mayor, Chile *Unveiling the G4-PAMAM capacity to bind and protect Ang-(1-7) bioactive peptide by MD simulations.*	Aug 2022
Molecular Biophysics Online Symposium, Universitätsklinikum Jena, Germany *Unveiling the G4-PAMAM capacity to bind and protect Ang-(1-7) bioactive peptide by MD simulations.*	Jun 2022
Symposium: Advances and Perspectives of the Nanoworld, FESI and CINVESTAV, Mexico *Unveiling nanocarrier/bioactive-peptide interactions: the G4-PAMAM/Ang-(1-7) case study.*	Mar 2022
German Workshop on Structural Predictions of Membrane Proteins, Forschungszentrum Jülich, Germany *Insights on anion selectivity across CLC antiporters by molecular simulations.*	Nov 2019
Dynapeutics International Summer School, University of the Basque Country, San Sebastián, Spain *Insights on anion selectivity across CLC antiporters by molecular simulations.*	Sep 2019
8th Molecular Dynamics Workshop, Campus Morelos UNAM, Mexico *In silico design of peptides to target resistin.*	Aug 2018
First Latin American Workshop and Conference on Systems Biology, CINVESTAV Zacatenco, CDMX, Mexico *In silico design of peptides to target resistin.*	May 2017

TEACHING AND MENTORSHIP

Co-Instructor, Biological Physics (PHYS 4380/5380) | Undergrad and graduate students from University of Idaho	Spring 2026
K–12 STEM Outreach Presenter, Scientist Day | Moscow Middle School, ID *Designed age-appropriate science activities for small student groups.*	Jan 2026
Mentor of undergraduate and graduate students | University of Idaho	2023 – Present
Undergraduate course on basic principles of MD simulations; mentorship of 1 undergraduate and 1 PhD student | ESM, IPN, Mexico	2021 – 2023
Teacher, Physics and Chemistry for Open High School | Echole Johnson Institute, Mexico	
Teaching Assistant, Quantum Mechanics I and Modern Physics | MSc in Applied Physics, CINVESTAV, Mexico	

INTERNATIONAL EXPERIENCE

Visiting Scientist | IAS-5/INM-9 Computational Biomedicine, Forschungszentrum Jülich, Germany *CLC ion channels.*	10/2019 – 02/2020

DAAD Visiting Scientist | IAS-5/INM-9 Computational Biomedicine, Forschungszentrum Jülich, Germany *CLC ion channels.*	05/2019 – 09/2019
Visiting Student, MEXFITEC Program | 8th class of Mexican student engineers in France	08/2009 – 07/2010
Internship | TLV Healthcare, France *Designer and manufacturer of hospital lighting units and equipment.*	01/2010 – 02/2010
French language and culture training | CAVILAM, Vichy, France	07/2009 – 08/2009

SCIENCE COMMUNICATION

- **Blogger,** Biophysical Society 2026 Annual Meeting. https://www.biophysics.org/blog/meet-the-2026-annual-meeting-bloggers
- **Writer,** Voices of Academia: "Challenges as a First-Generation Student: Studying Abroad" (2024). https://voicesofacademia.com/
- **Writer,** Avance y Perspectiva, CINVESTAV (July 2021): "Diseño de péptidos para aplicaciones biomédicas." https://avanceyperspectiva.cinvestav.mx/diseno-de-peptidos-para-aplicaciones-biomedicas/
- **Talk,** "Ciencia computacional de péptidos cortos: aplicaciones, fronteras y perspectivas." https://www.facebook.com/BiocaMol/videos/1013668386690543
- **Talk,** "Estudios computacionales moleculares," Radio IPN. https://www.facebook.com/RadioIPNOficial/photos/a.199848847434895/1349046552515113/

PROFESSIONAL ASSOCIATIONS

American Chemical Society (ACS)	09/2026 – Present
Biophysical Society (BPS)	12/2023 – 2026
National Postdoctoral Association	09/2023 – Present
Thematic Network of Structure, Function and Evolution of Proteins (REFEP)	09/2023 – Present
Society for Industrial and Applied Mathematics (SIAM)	06/2024 – 2025

PEER REVIEW AND SERVICE

Postdoc Selection Committee | University of Idaho	Oct 2024
Reviewer, Frontiers in Computer Science	Jul 2024
Service, 78th Northwest Regional Meeting (NORM)	Jun 2024
Reviewer, ChemistrySelect	Aug 2023
Reviewer, New Journal of Chemistry	Apr 2023
Reviewer, Food Science and Biotechnology	Apr 2021

WORKSHOPS AND TRAINING

Computer Vision Symposium | University of Idaho *AI-based computer vision.*	Oct 2023
Falcon Supercomputer Workshop | University of Idaho *Logging in and running jobs.*	Apr 2023
Characterization Techniques of Biomaterials and Biological Samples | UPIIG-IPN, Mexico *AFM technique.*	Mar 2021
Molecular Bioinformatics | RWTH Aachen, Germany *Sequence alignments and homology modeling.*	Sep 2019
Minicourse in Protein Physics (Paolo Carloni) | REFEP, Mexico *Connection between molecular simulation and systems neurobiology.*	Nov–Dec 2017
Workshop NVIDIA–IBM | ABACUS, CINVESTAV, Mexico *GPU programming with OpenACC, image classification with DIGITS, PowerAI, and AI.*	Nov 2017